

Windows CLI & Git Cheat Sheet

Quick Reference Guide for Developers

1. Navigating the Filesystem

Command	What it does	Real-world Equivalent
<code>cd <folder_name></code>	Change directory	Double-clicking a folder
<code>cd ..</code>	Move up one level	Clicking the "Back" or "Up" arrow
<code>dir</code>	List files and folders	Opening a folder to see what is inside
<code>mkdir <folder_name></code>	Make a new directory	Right-click → New → Folder

2. File Operations

Command	What it does	Pro-Tip / Warning
<code>type NUL > filename.ext</code>	Creates a new empty file	Great for spinning up a quick script or text file
<code>copy <source> <dest></code>	Copies a file	<code>Ctrl + C</code> then <code>Ctrl + V</code>
<code>move <source> <dest></code>	Moves a file (or renames it)	<code>Ctrl + X</code> then <code>Ctrl + V</code>
<code>del <filename></code>	Deletes a file	Permanently deletes it (skips the Recycle Bin!)
<code>rmdir /s /q <folder></code>	Force deletes a folder	<code>/s</code> deletes subfolders, <code>/q</code> quiet mode (no prompts)

3. System & Network Troubleshooting

Command	What it does	When to use it
<code>ipconfig</code>	Shows IP addresses	Checking your local IP address
<code>ping <host/IP></code>	Sends packets to an address	Checking if a server or website is alive
<code>netstat -ano</code>	Lists active network connections	Essential for when code says "Port already in use"
<code>taskkill /F /PID <id></code>	Force kills a process by ID	Stopping a rogue Python or Java background process

4. The Git Core Toolkit

Command	What it does	The "Under the Hood" Reality
<code>git init</code>	Turns folder into a Git repo	Creates a hidden <code>.git</code> folder to track history
<code>git clone <url></code>	Downloads an existing repo	Grabs a project from GitHub locally
<code>git status</code>	Shows modified/staged files	Your ultimate safety check before you commit anything
<code>git add <filename></code>	Stages a file	Saying: "Hey Git, look at these specific changes"
<code>git add .</code>	Stages <i>all</i> modified files	Shortcut, just check <code>git status</code> first to avoid adding junk
<code>git commit -m "msg"</code>	Saves staged changes	Creates a permanent checkpoint in your code's timeline
<code>git push origin <branch></code>	Sends commits to remote	Uploading your checkpoints to GitHub
<code>git log --oneline</code>	Shows clean commit history	Reading the summary of how your project evolved